

Linux Programming: Assignment-2:

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SECTION- C

1.What does the command pwd, whoami, and hostname display?

Ans

Commands:

- 1) Pwd: displays path of current directory
/home/user/document
- 2) Whoami: displays current username
jyothsna
- 3)hostname: displays name of the system or computer
Ubuntu

2.Write the command to create a directory named “project” inside the /home/student folder and keep three .txt file into it. Give output snapshot.

Ans

`mkdir -p /home/student/project`

`touch /home/student/project/file1.txt /home/student/project/file2.txt /home/student/project/file3.txt`

Mkdir: creates project dictionary

Touch: creates three text file that are empty

Output snapshot:

```
$ mkdir -p /home/student/project
$ touch /home/student/project/file1.txt
/home/student/project/file2.txt /home/student/project/file3.txt
$ ls /home/student/project
file1.txt file2.txt file3.txt
```

3.Explain the difference between absolute path and relative path with proper examples.

Ans

Absolute path: this path starts from root directory ,starts from “/”
it also shows full location of file works anywhere in the system
Example: /home/student/project/file.txt

Relative path: this path is relative to current directory
It depends on pwd
Doesnt start with”/”
Example:project/file.txt

4.What command will give you the already executed command traces in the terminal. Give output snapshot.

Command:

history: give us all the used commands

Output:

```
$ history
1  pwd
2  whoami
3  mkdir -p /home/student/project
4  touch /home/student/project/file1.txt
5  ls /home/student/project
```

```
6 cd /home/student/project
7 history
```

5. Compare the working functionality of find and locate command. Which one is faster and why?

Ans

Command:

find: The “find” command searches for files in realtime by traversing in the directory tree. It scans the filesystem directly. This means it is always accurate and up-to-date. It is also very flexible allows conditions like size, because it performs live search it can be slow for larger files

locate: locate command uses a prebuilt database to find files. the results from locate can be outdated if the database hasn't been updated recently. Unlike find, locate is mostly limited to simple name-based searches.

6. Which command is used to modify file permissions in Linux? Give an example

Command:

Chmod: short for change mode is used to change the file permission

Example

```
chmod u=rwx,g=rx,o=r example.txt
```

Here the u is given permission to read, write and execute

While g can only read and execute the file and o can only read the file

7. A file has permissions -rw -r- -r- -. What does this mean?

Ans

Here the first "-" means that it is a regular file and not a directory

The next character "rw-" means this is a owner's permission to read and write in this file

The "r- -" means this are group's permission to only read the file

The next characters are "r- -" are given to others and the file can only be read by them

8. Explain the difference between chown and chgrp with an example

Commands

chown: this command changes the owner of the file or directory

It can also change group if the time is specified

Only rootusers can change the ownership

Chgrp: changes ownership of the file

Only changes the group and not the owner

Example

chwon:Example: chown user1 file1.txt

chrgrp:chgrp students file1.txt

9.A file needs to be accessible by multiple users but only writable by the owner. How will you set permissions?

Ans

command:chmod 644 filename

This means that the owner can only read or write the text file and the others can only read

10.How do you check the manual page for any Linux commands?

Command:

`man <command>`

This command is shortened for manual that means it has all the syntax, usage guide for all the command in linux